



Assignment 01

Data Analysis and Visualization

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Football Dataset

The hypothesis for this data set was that germany would have the highest total goals. This was graphed to be tested. Then the hypothesis was rejected as this wasn't the case.

Basic Exploration

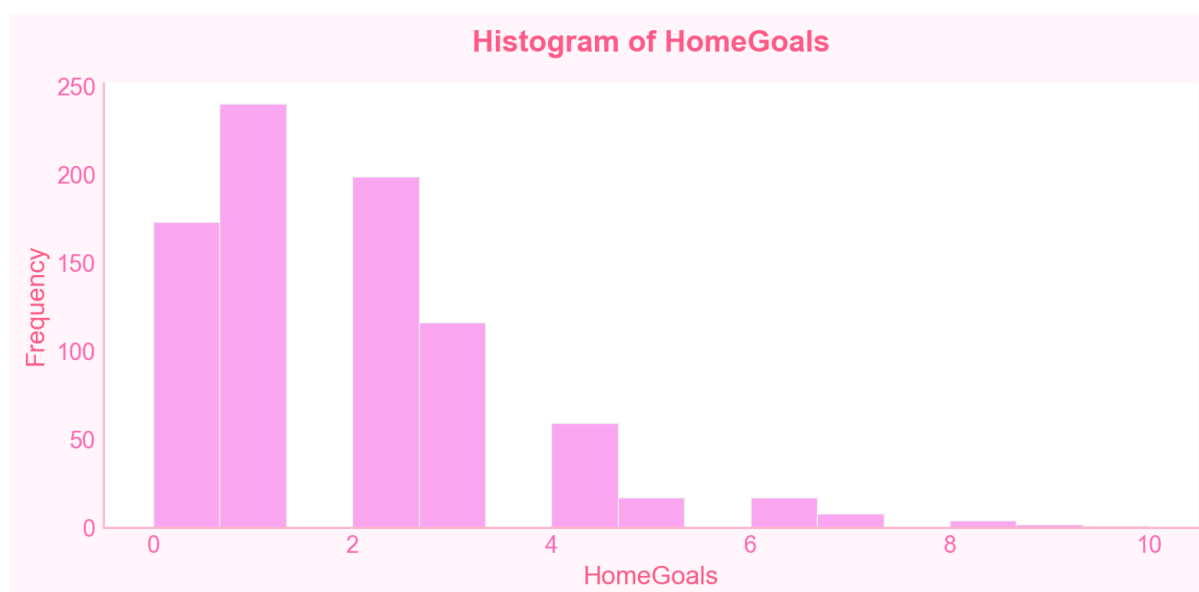
Explored data by looking at its head and tail. The dataset had 852 rows and 12 columns. Date was datetime, Year, Home Goals and Away Goals were numerical and the rest were objects. There were no apparent missing values, but there were blank spaces in Observations. There were 16 duplicate rows. Explored summary statistics using `df.describe()`.

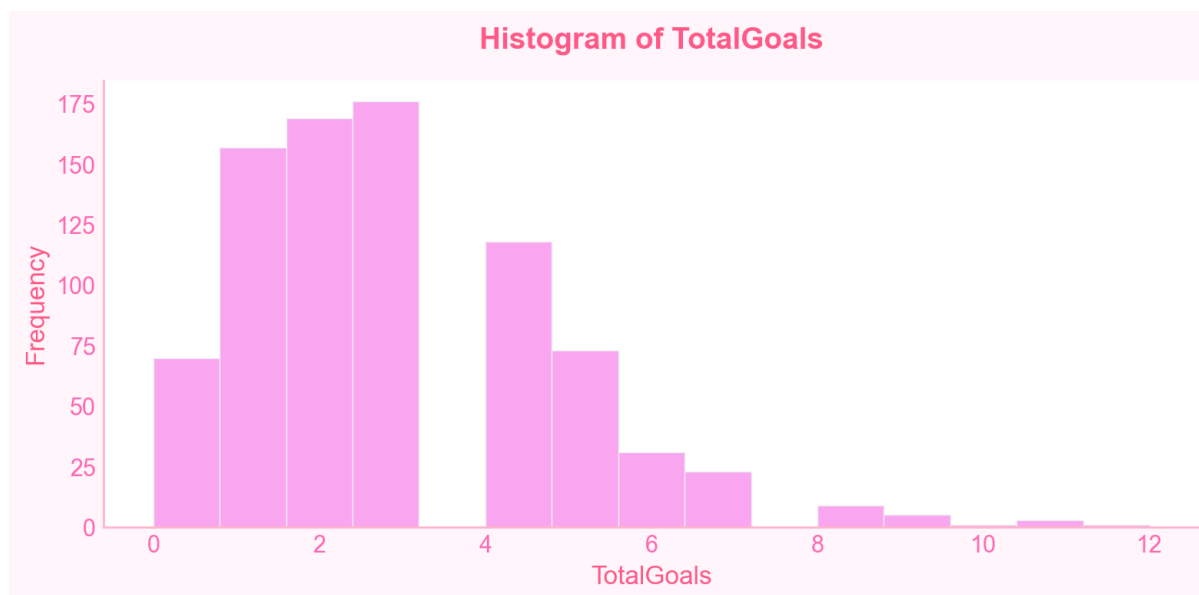
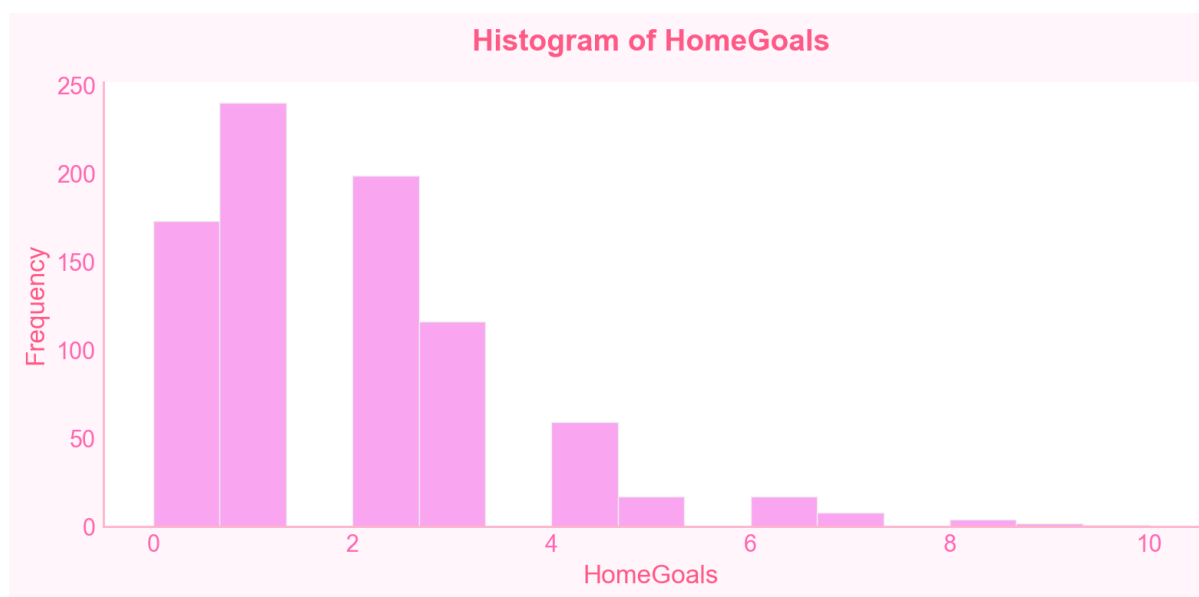
Data Preprocessing

Changed column names to standard names. Like changing both 'USA' and 'United States of America' to 'United States', for data consistency. Corrected data types and removed duplicates and blank strings. Made a new column called Total Goals.

Univariate Analysis

Histograms

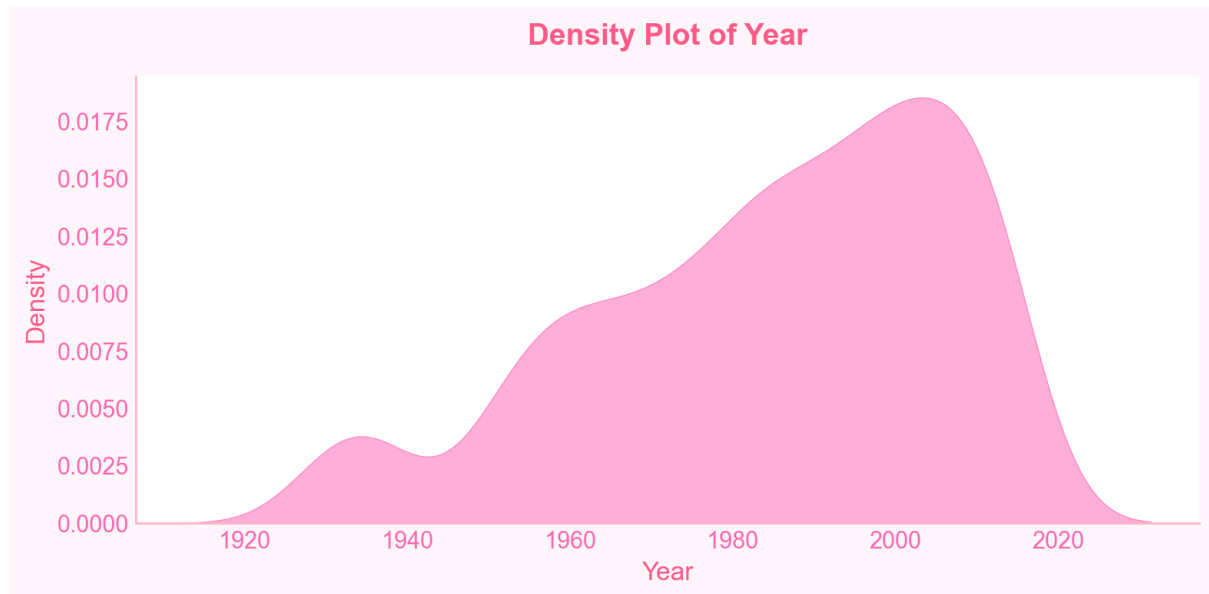




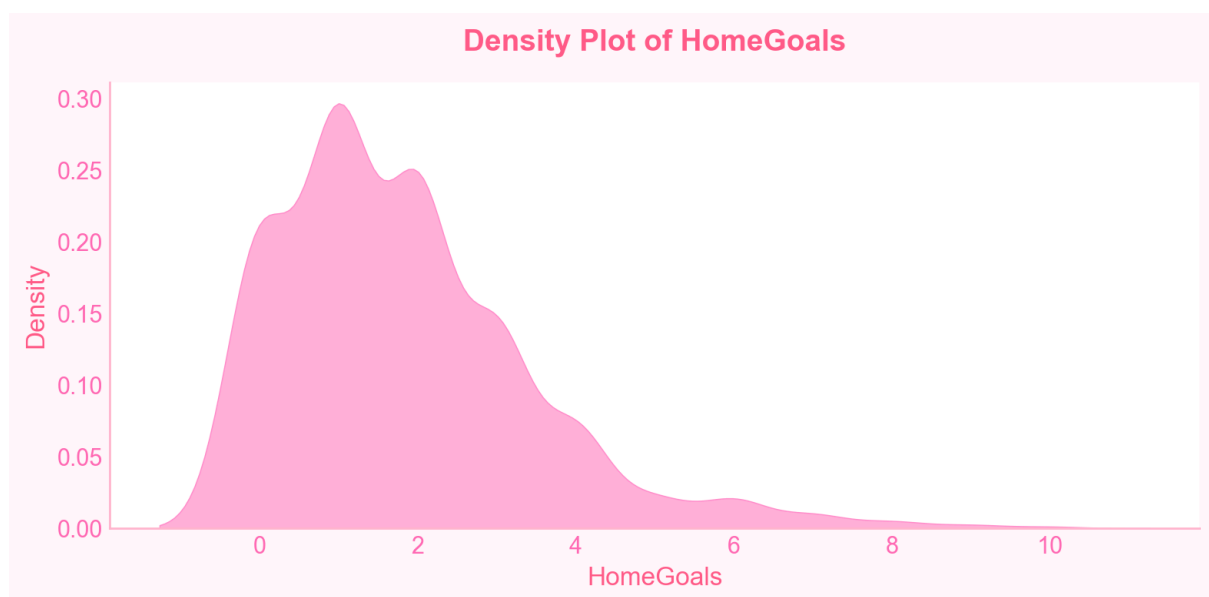
Made histograms for all numerical columns. Both Home goals and Away Goals were negatively skewed with the mode being 1. Total Goals were also slightly negatively skewed .



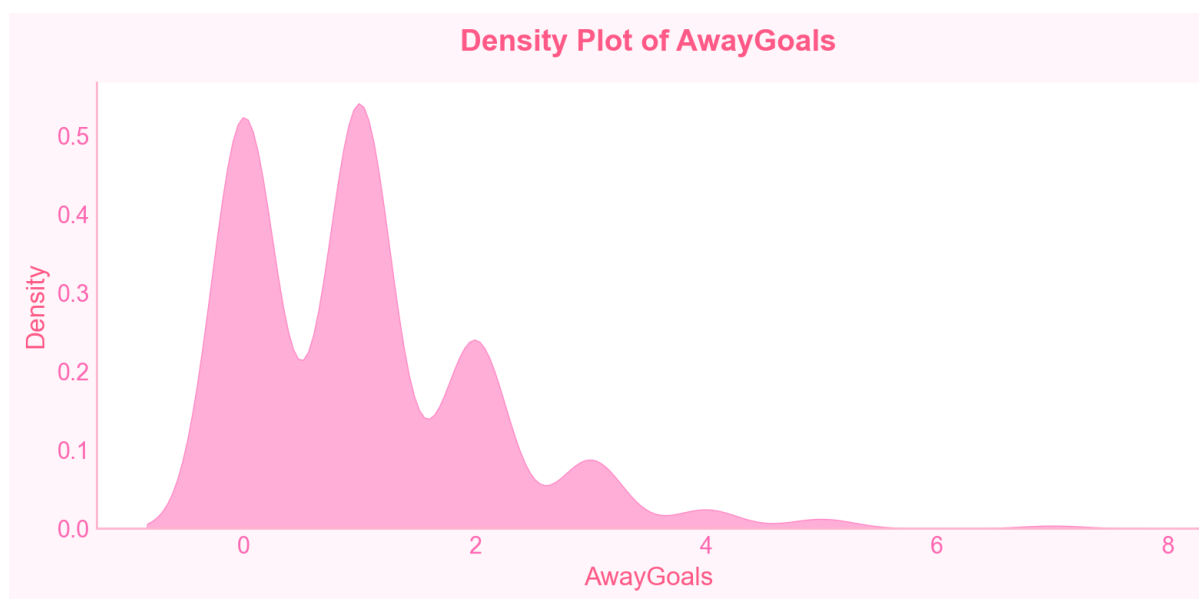
Density Plots



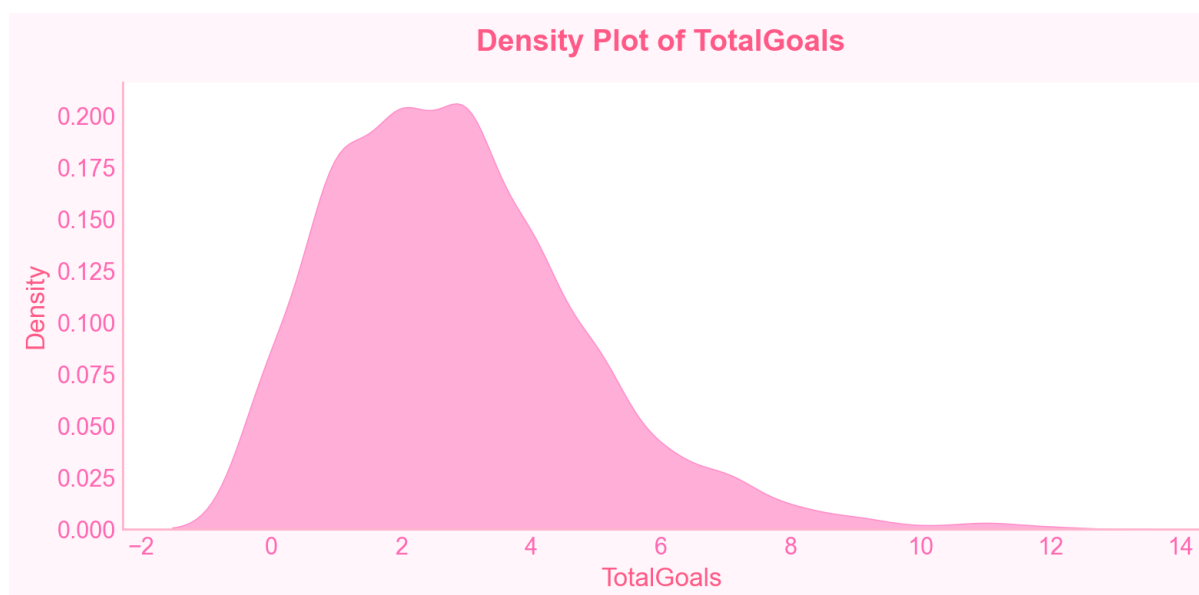
The density plot of year showed an expected increase with time.



Home goals had a negative skew of a normal distribution.



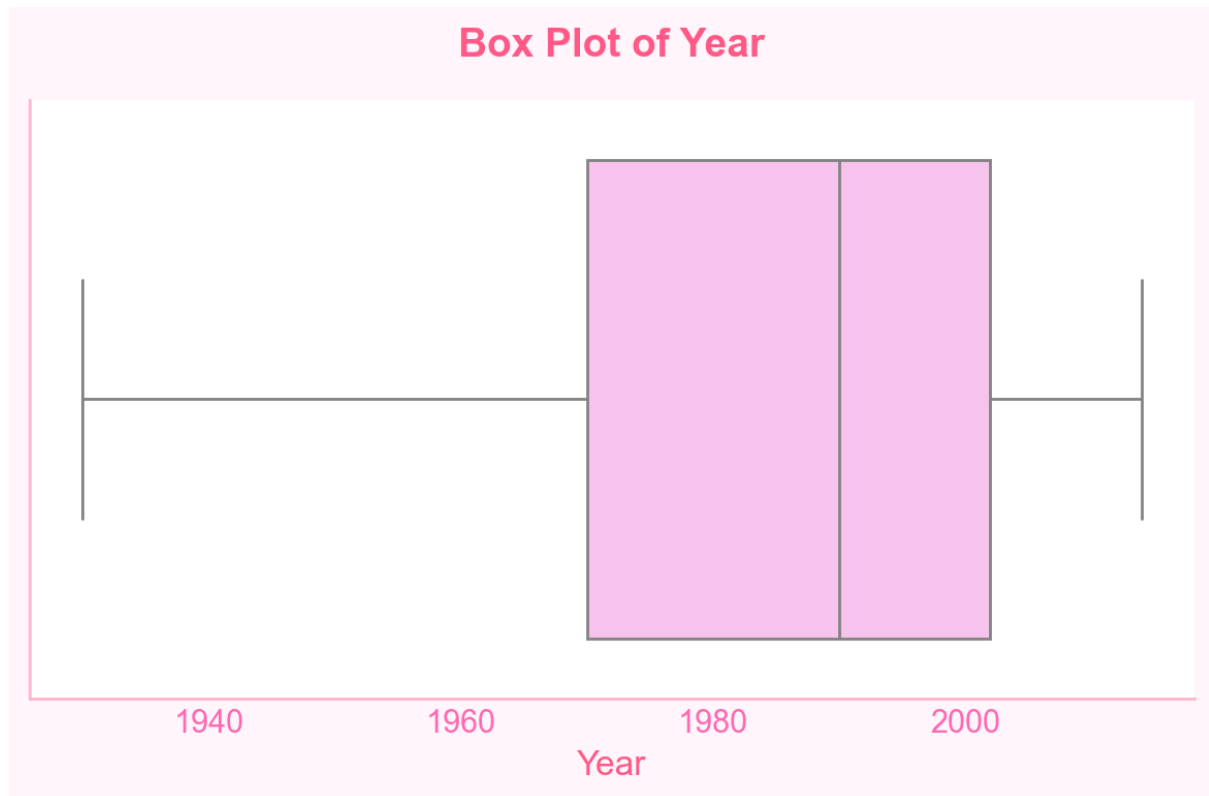
Away goals was not a normal distribution but was also negatively skewed.



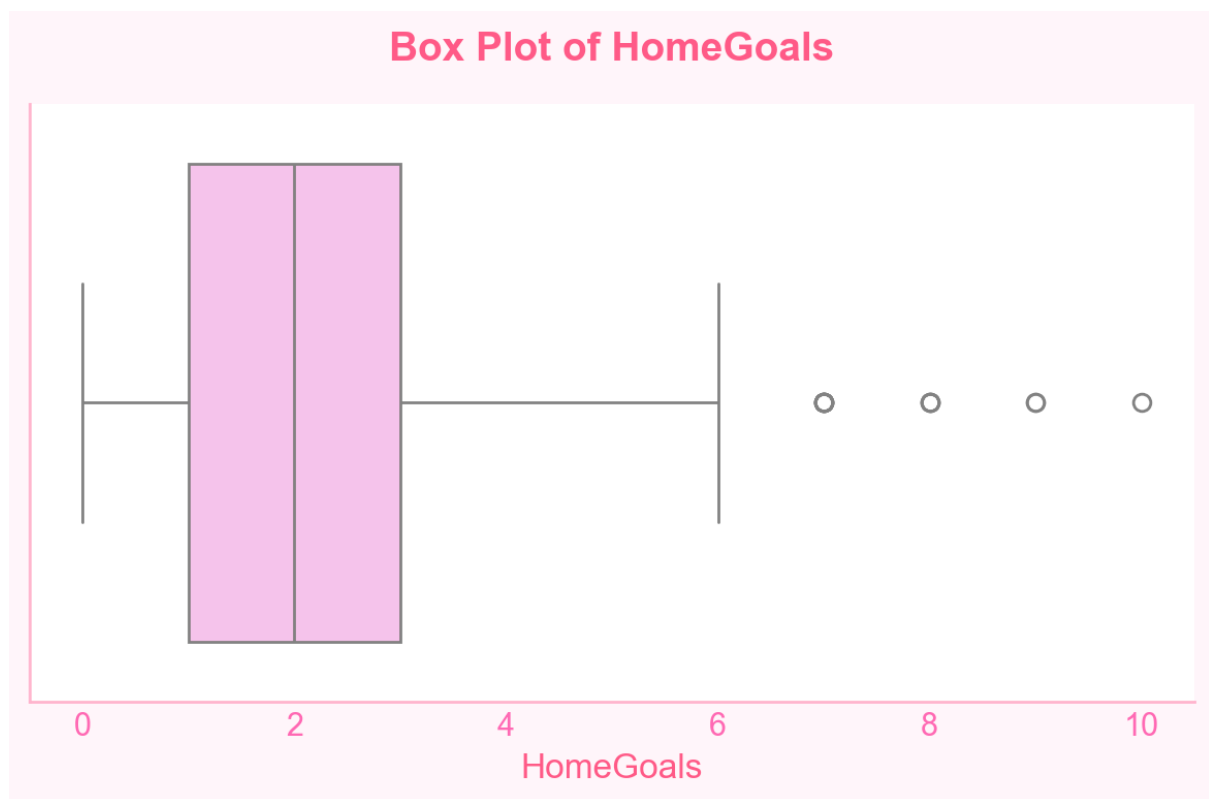
Total goals was a slightly skewed normal distribution curve, showing how total goals are evenly distributed.



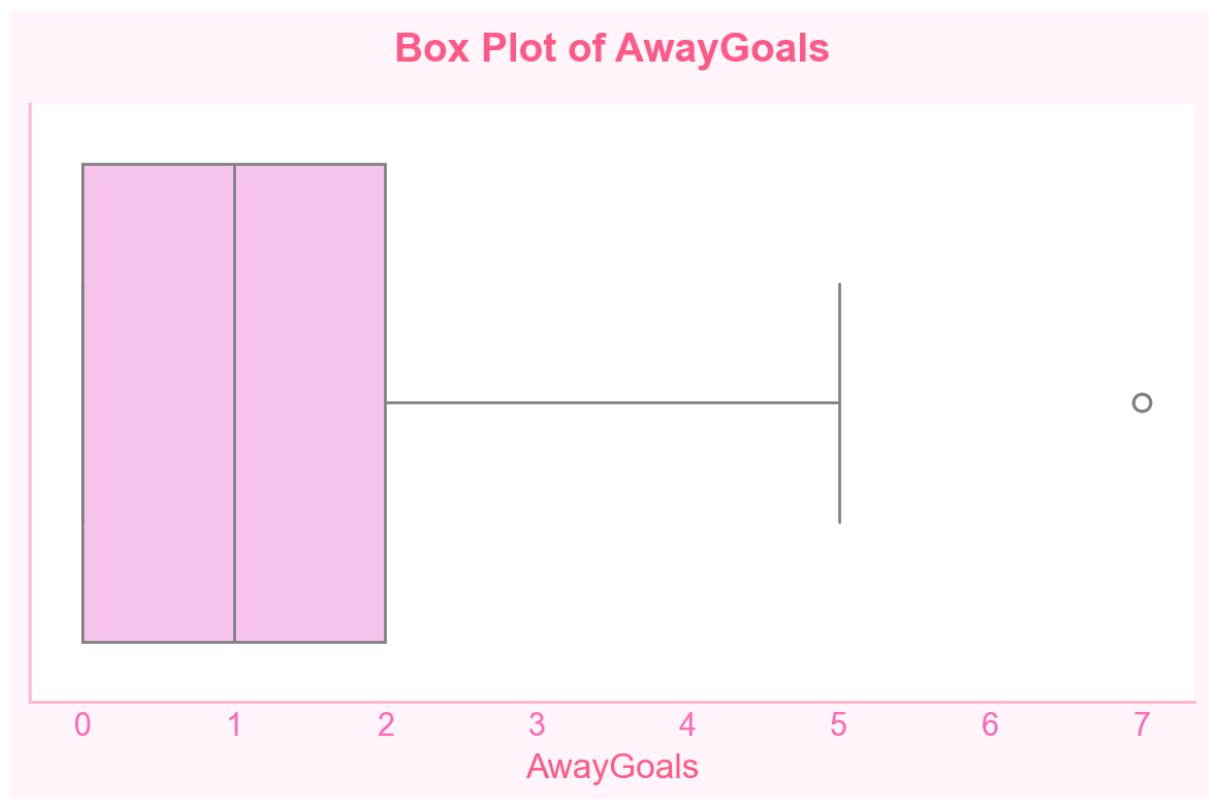
Boxplots



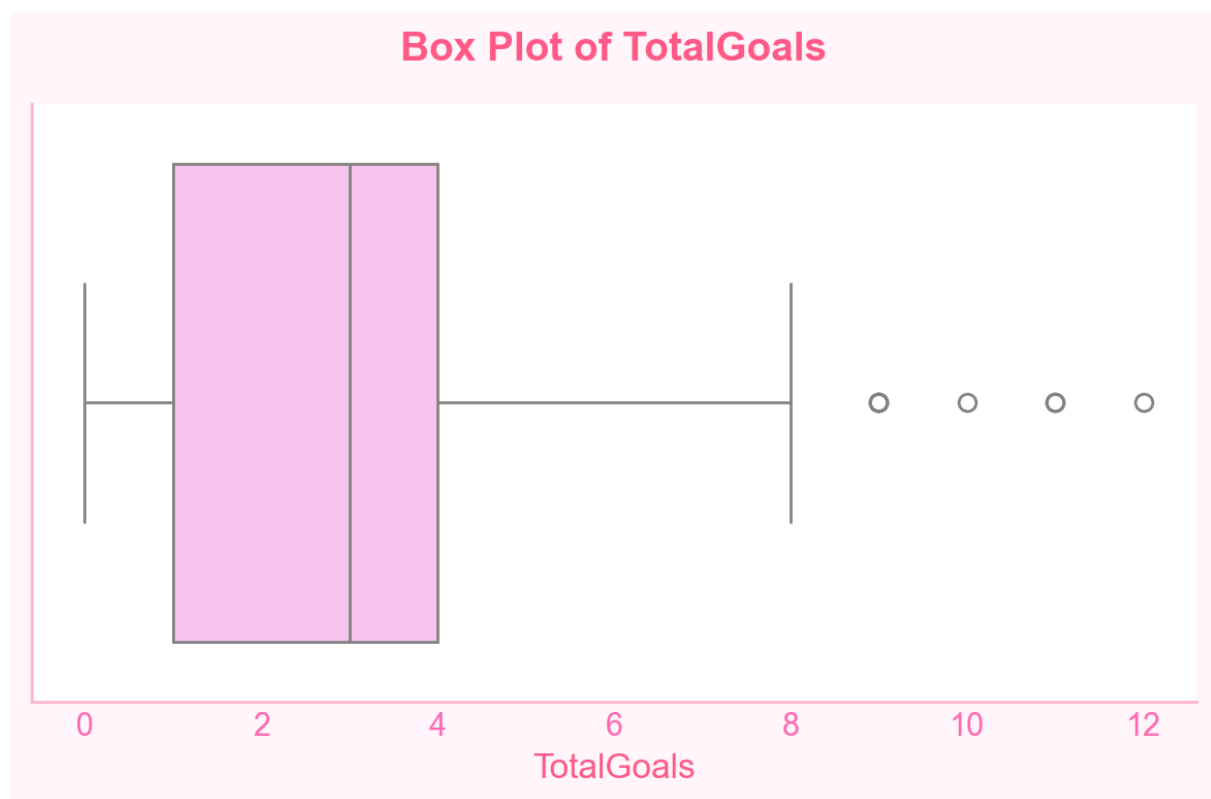
Years had no outliers and were positively skewed.



Home goals had the median 2 and 4 outliers of goals 7, 8, 9,10.



The values of away goals were less than home goals. With the median 1 and only one outlier of 7 goals.



Total goals had a median between 3 and 4 with 4 outliers 9,10,11 and 12 total goals.



Bivariate Analysis

Scatter plots



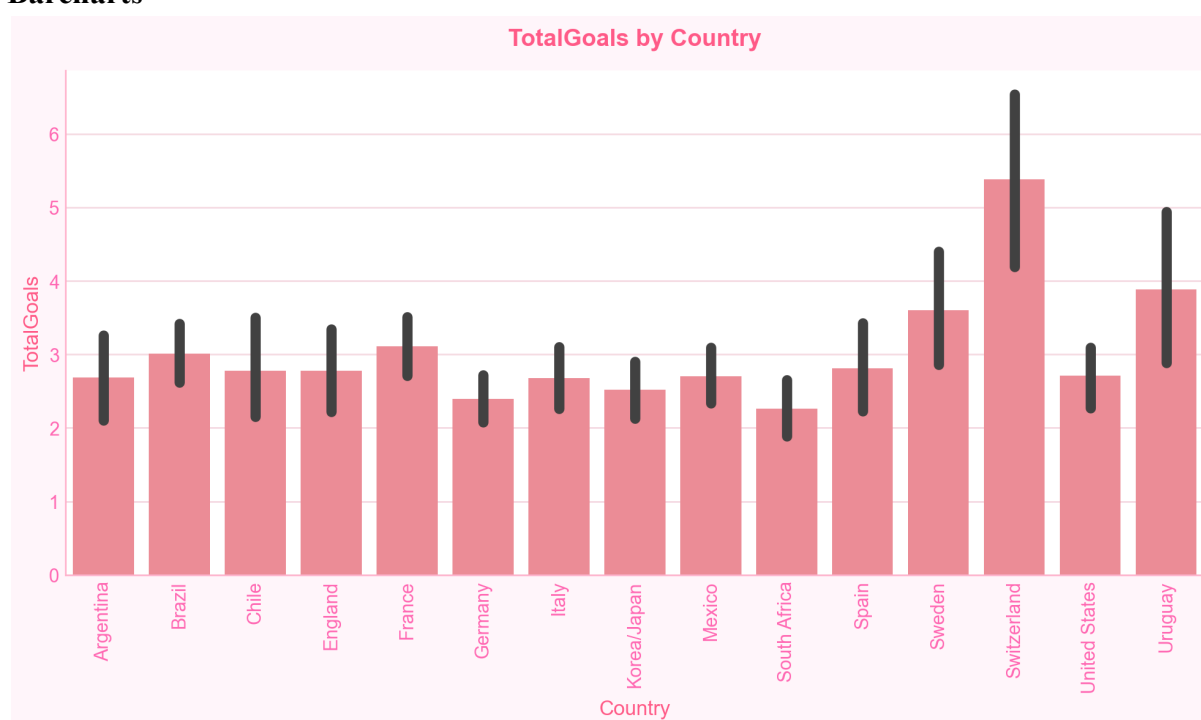


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Correlation Matrix

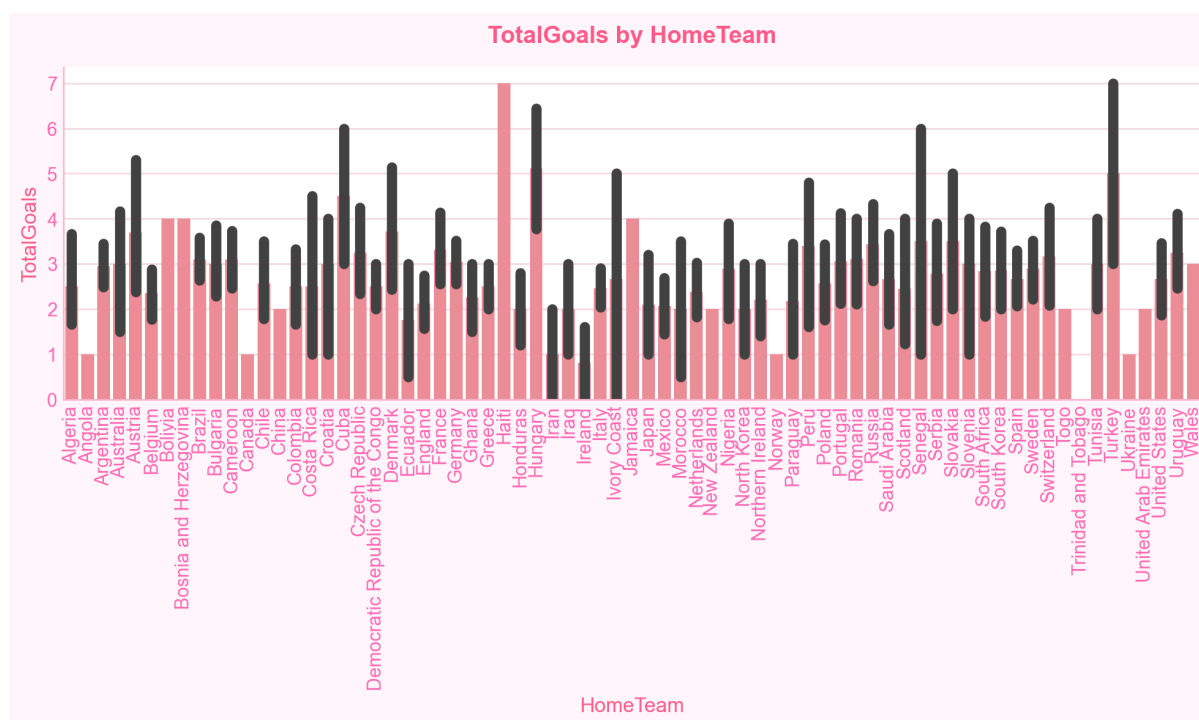
The correlation matrix showed the correlations between all numeric variables between 1 and -1. 1 being a highly positive correlation and -1 being a highly negative correlation.

Barcharts



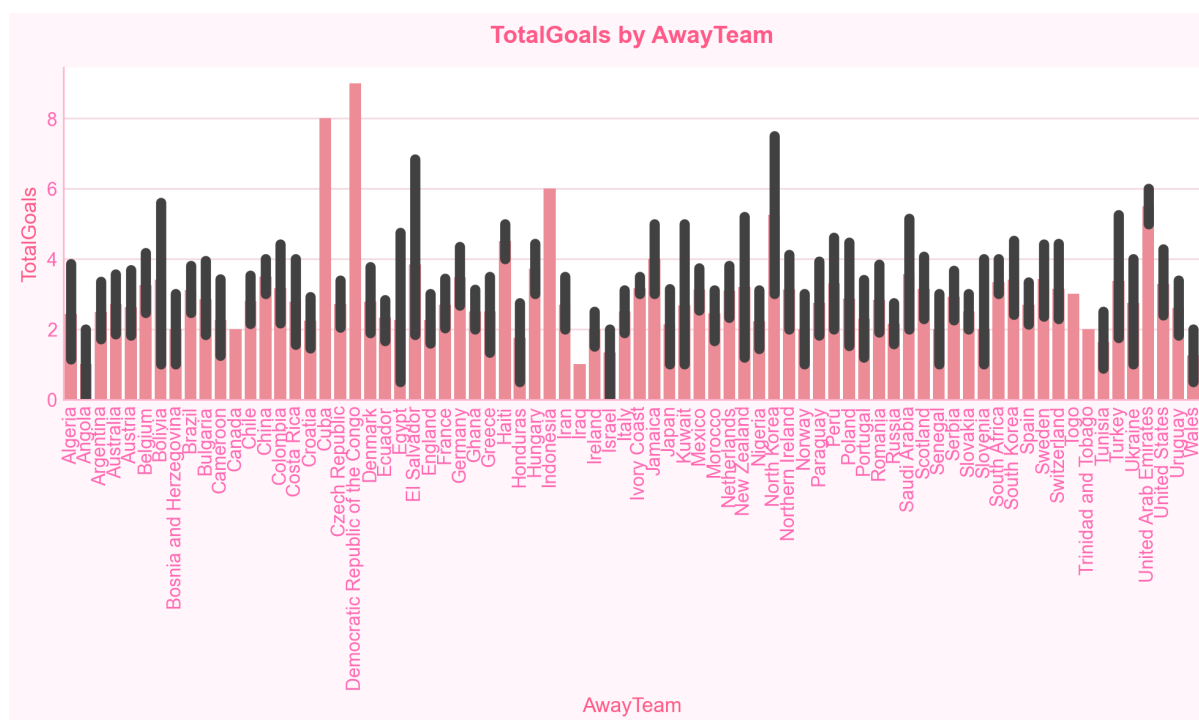
The scatter plots showed one target numerical column total goals compared to all categorical columns. The figure above shows all countries and their total goals across time. The graph shows that Switzerland has the most goals, not Germany. So the hypothesis was rejected.

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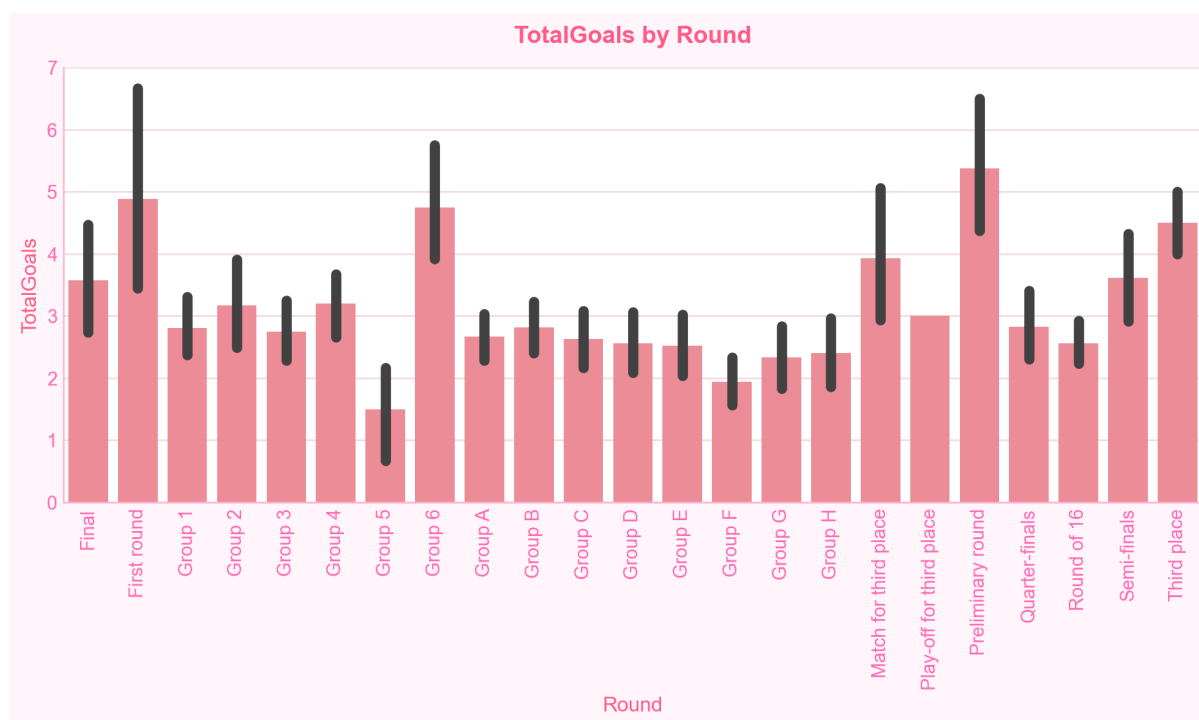


Compared to home teams Haiti has the most goals at 7 although the median seems to lie at around 2.

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Democratic Republic of Congo has the most goals at 9 as the away team and Iraq has the least at 1.



Compared by Round the most goals are scored in the first round and the least in Group 5.

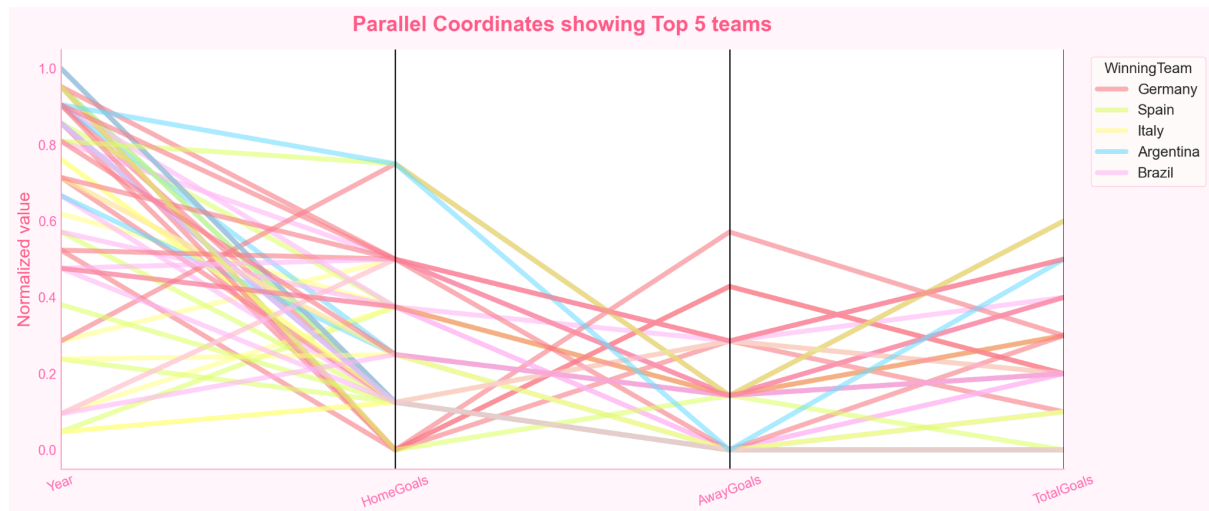
Multivariate Analysis

Scatterplot Matrix

Same thing as above.

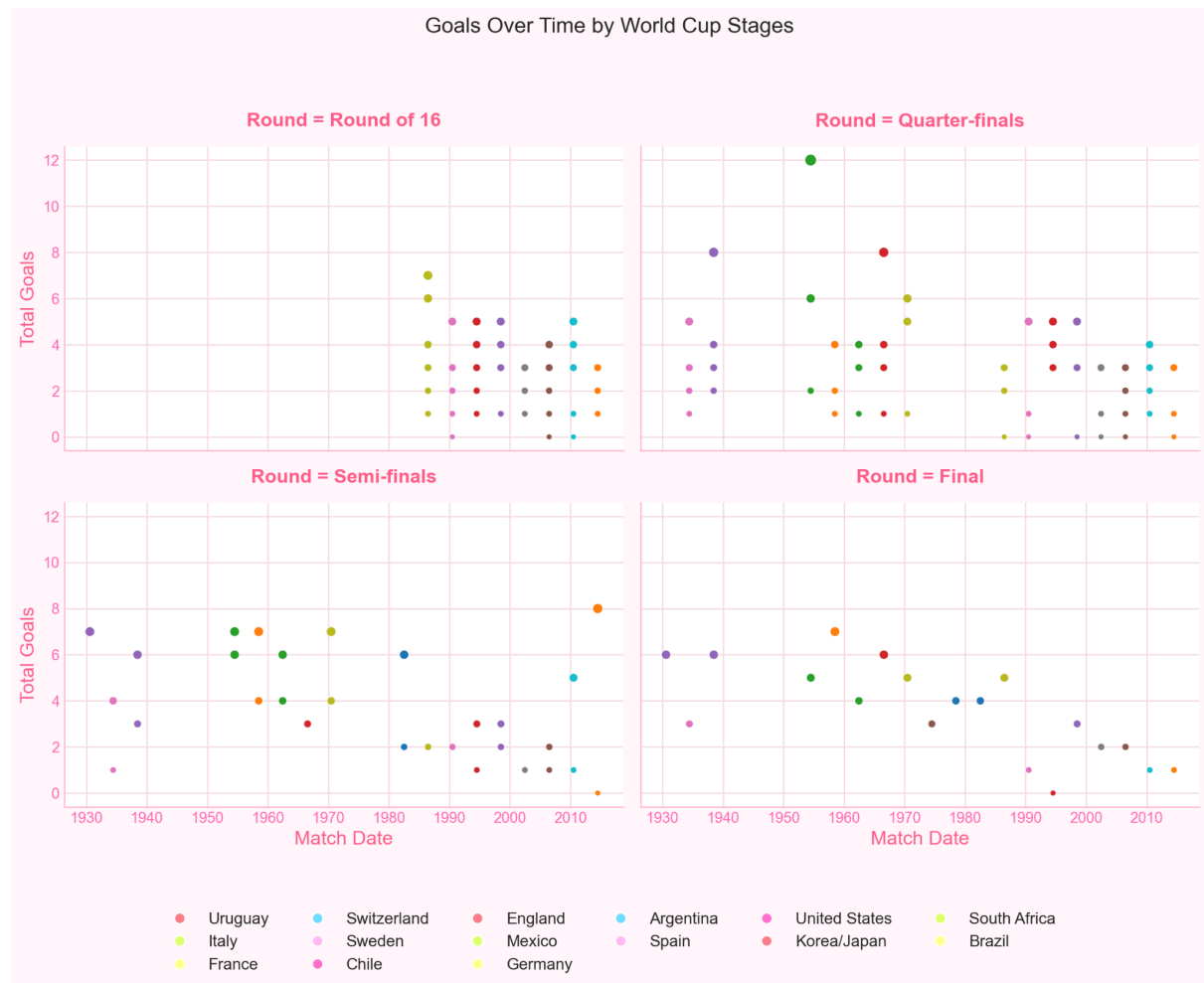


Parallel Coordinates



This graph shows the top 5 teams in number of goals scored compared to all numerical attributes. The axis has been normalized to be equal. It shows the positive correlation between total goals and away goals and a negative one between home goals and away goals.

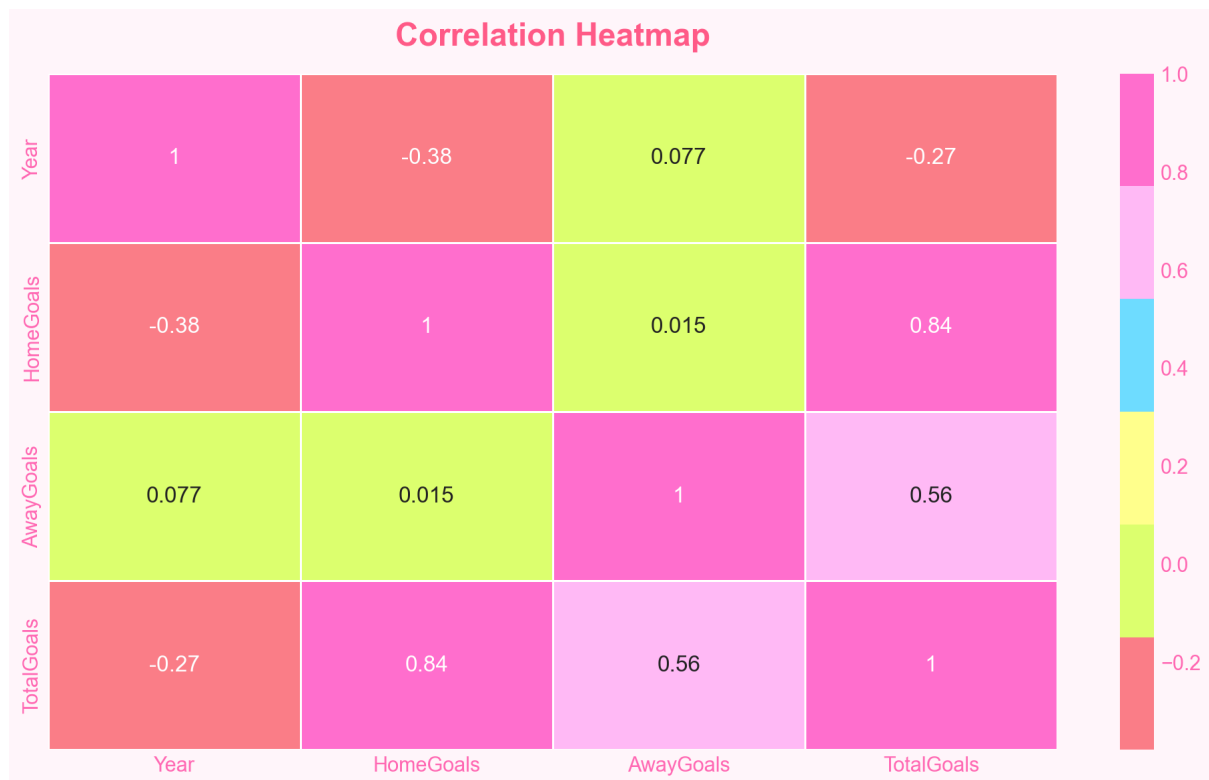
Plot Trellis Plot



This plot shows total goals scored in 4 rounds for all years in the data set for a set of 15 countries where the world cup was also hosted once.



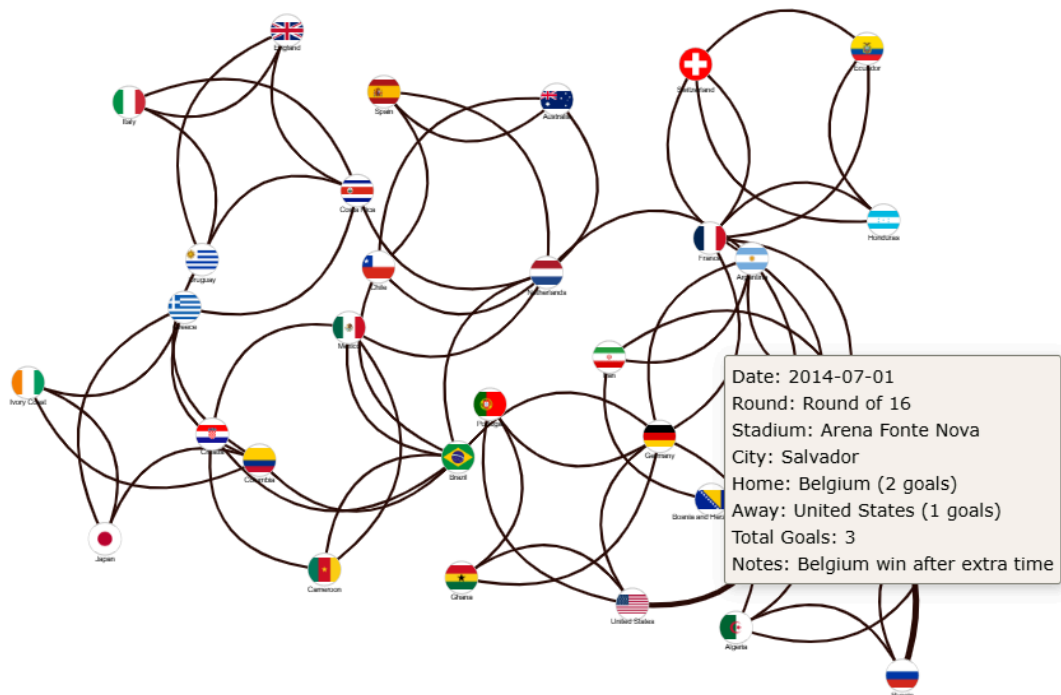
Heatmap



Heatmap of the correlational matrix above. It shows that home goals and total goals are highly positively correlated. However, away goals and year and home goals and year have no correlation.

Graph Network of all Teams in World Cup 2014

I first found all teams that played then mapped them to their country codes. Then found their flags online and used them as images for the graph nodes. Each edge had hover text showing the date it happened, the stadium and round name, any notes and the goals by both teams.



AirBnB Dataset

The hypothesis for this data set was that properties with higher prices would have higher ratings. This hypothesis was tested by graphing prices compared to different properties and their ratings. The hypothesis was then tested to be correct.

Basic Exploration

The airbnb data had 20478 rows and 13 columns. It was explored by looking at its head and tail, checking data types, missing, duplicate and unique values. Summary Statistics were used to get a clear overview of the data frame. There were 4 datatypes: int, float, string and datetime. 8 columns were numerical, 1 was datetime, and 3 were string. Some columns had missing values and there were duplicates as well.

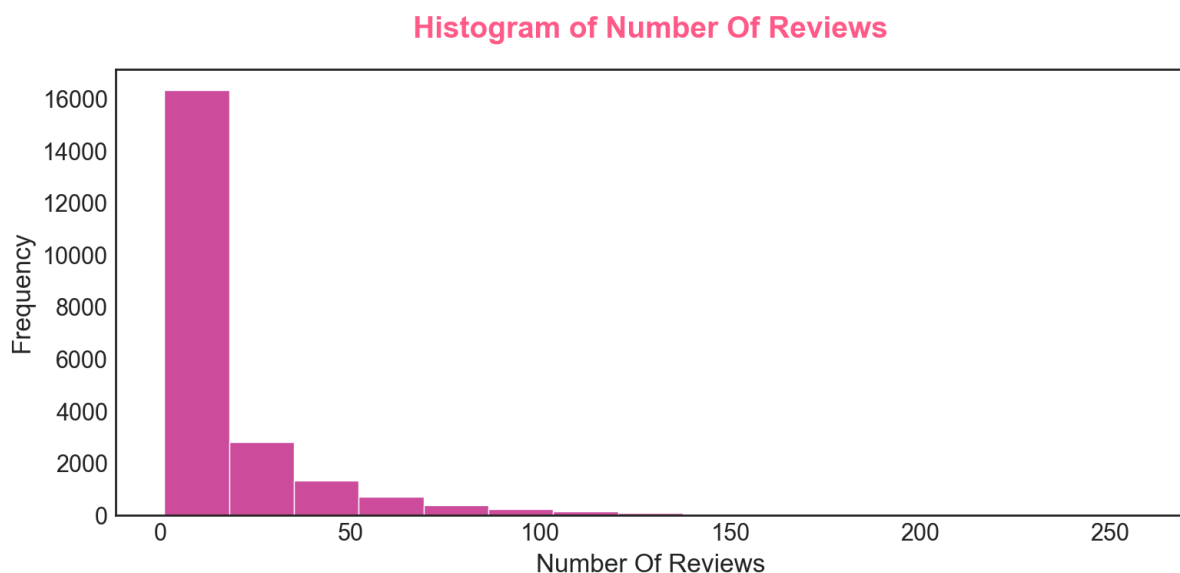


Data Preprocessing

One column name was changed, then the data types were corrected from object to string and to categorical. Missing values were removed and number of records was dropped as it had only 1 value. Duplicates were removed. New column price per bedroom was made.

Univariate Analysis

Histograms

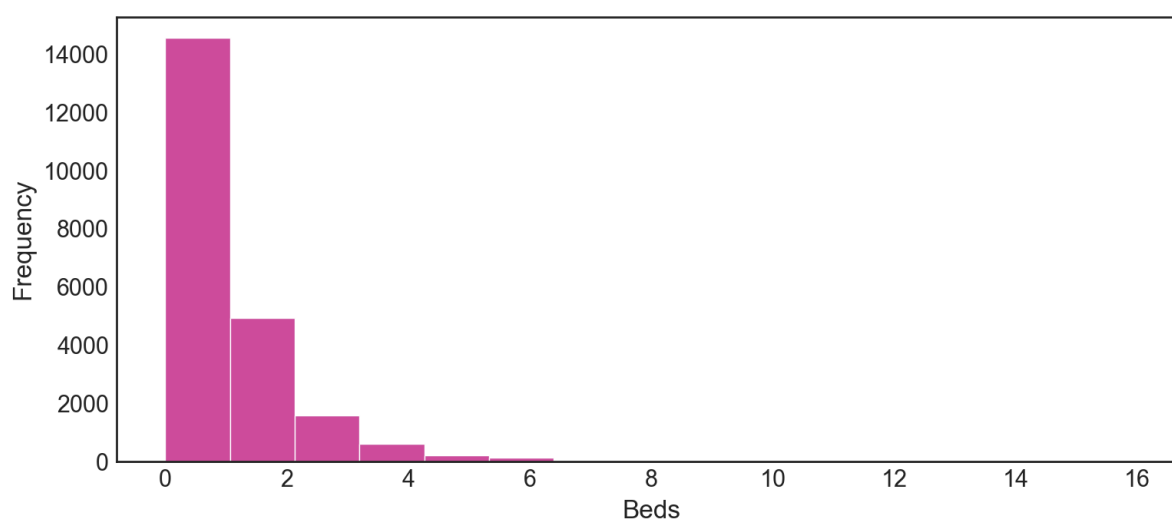


The number of reviews for an apartment were negatively skewed and were very close to zero on average.



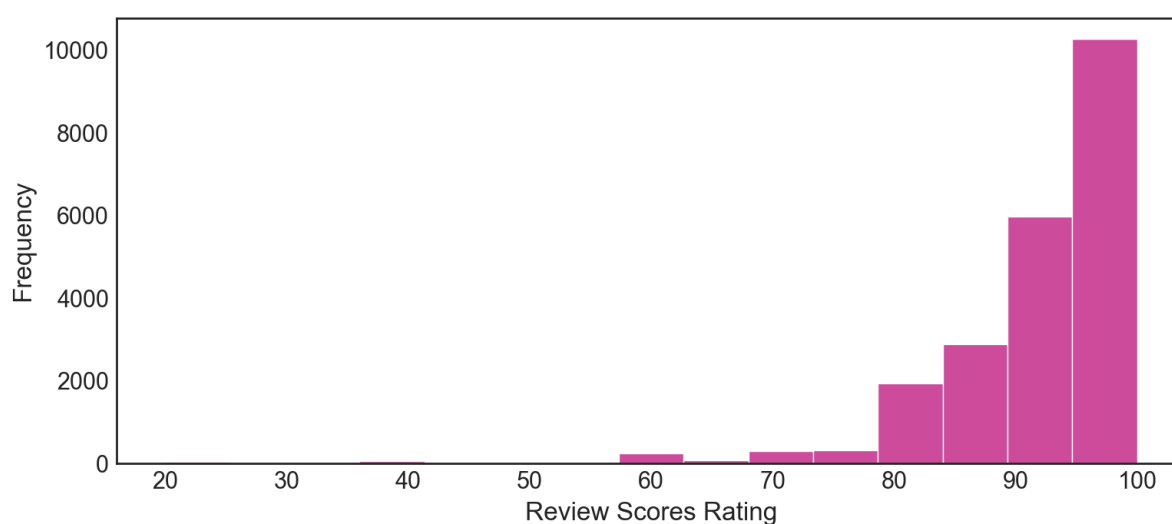
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Histogram of Beds



The number of bedrooms were also normally distributed and negatively skewed.

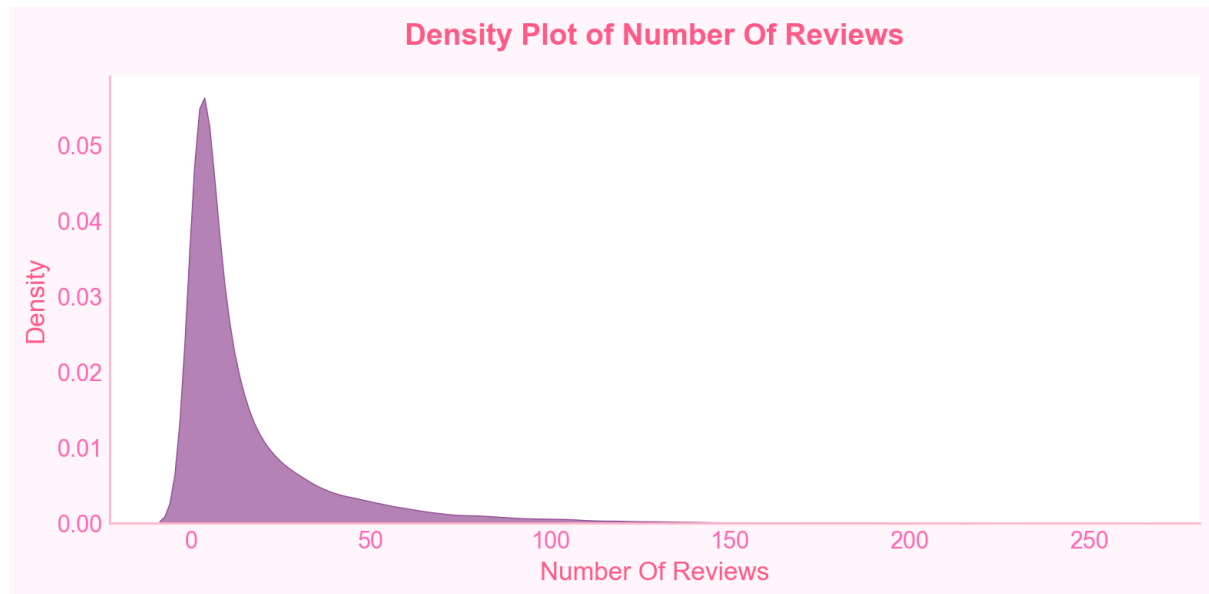
Histogram of Review Scores Rating



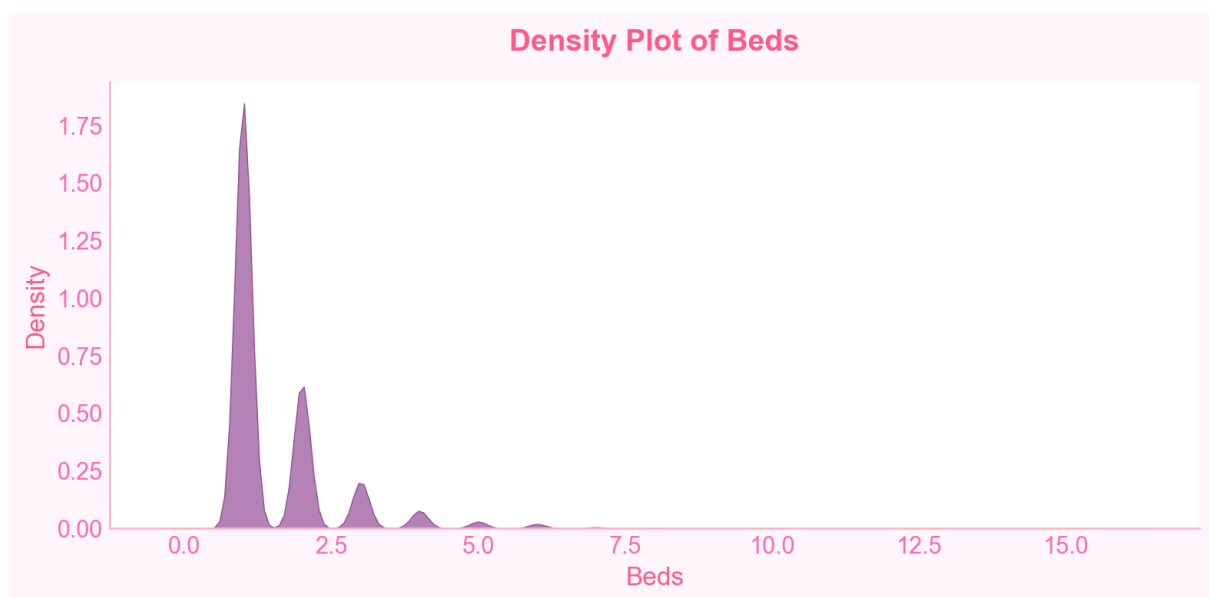
review Scores Rating were positively skewed as a positive curve with the highest frequency 10000 for the last bin 95-100. This shows that scores were very positive.

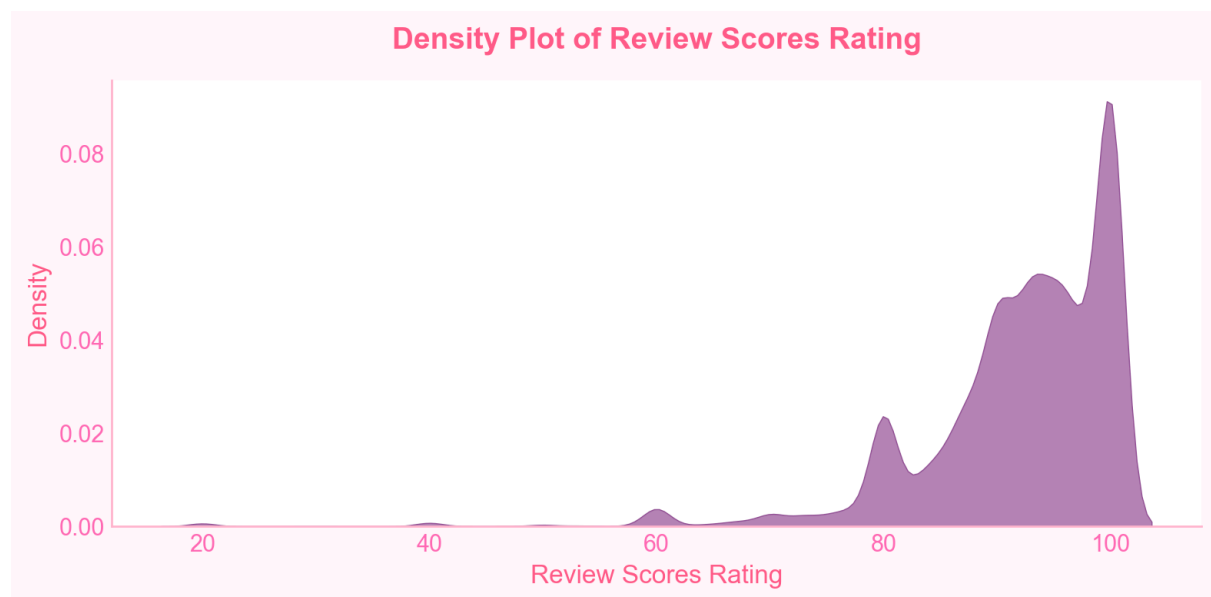


Density Plots



fdsa





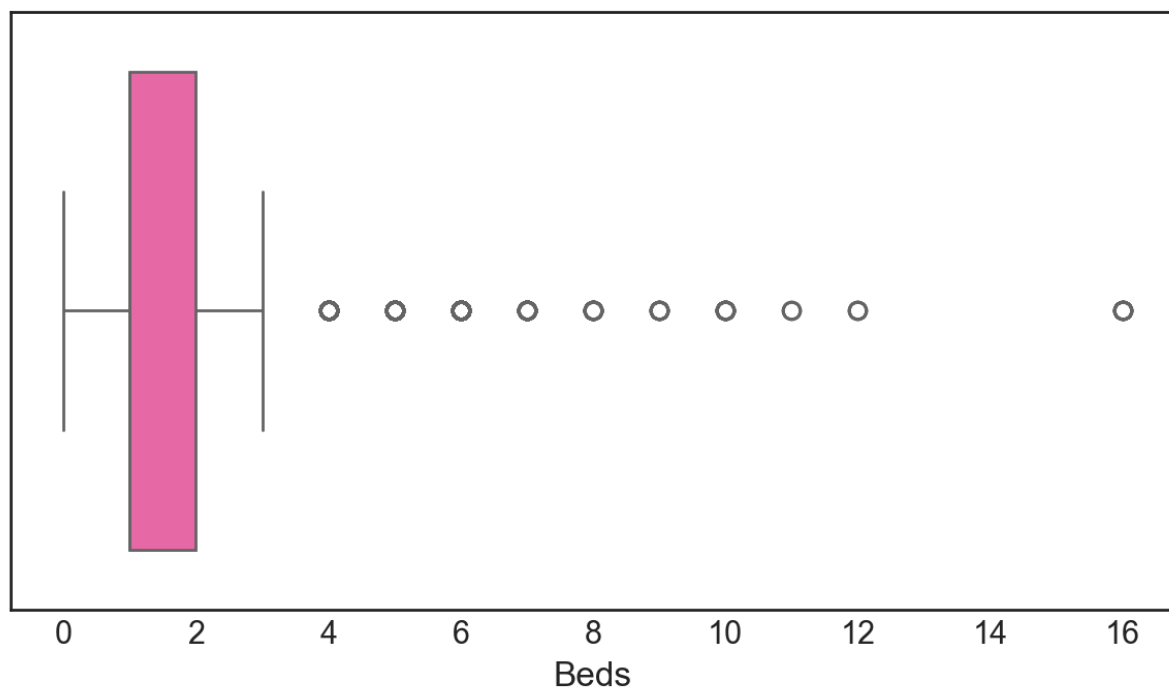
Boxplots





The number of reviews had a lot of outliers but was still negatively skewed.

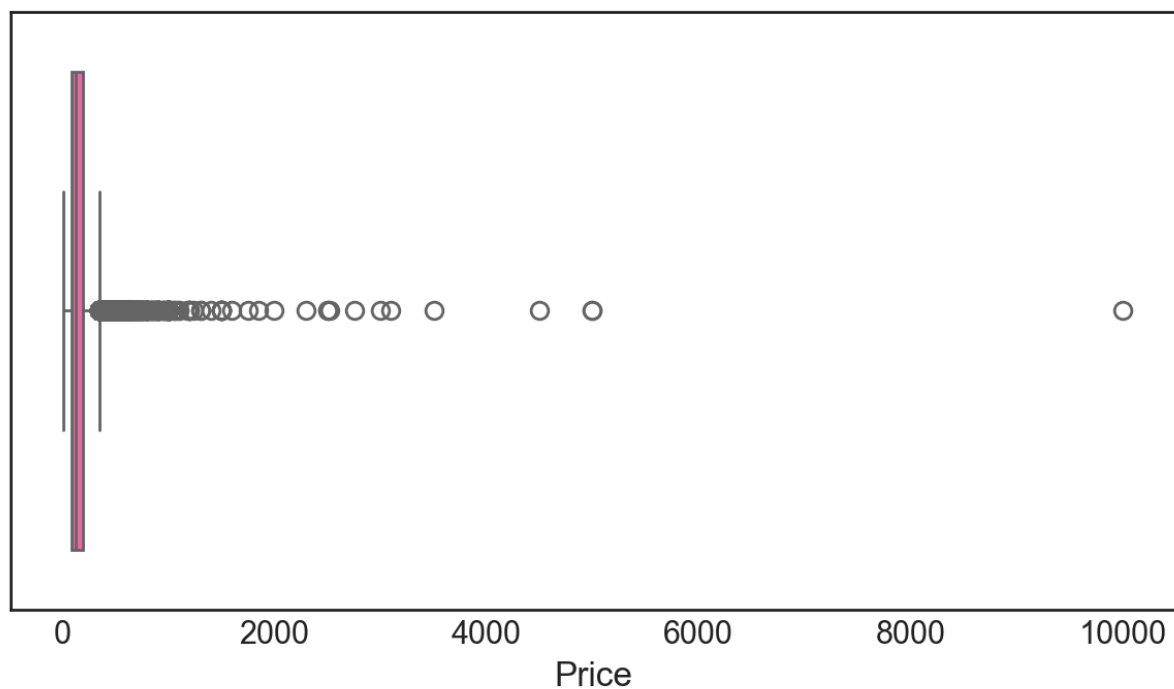
Box Plot of Beds



The number of bedrooms had a few outliers and its q1 was its median.



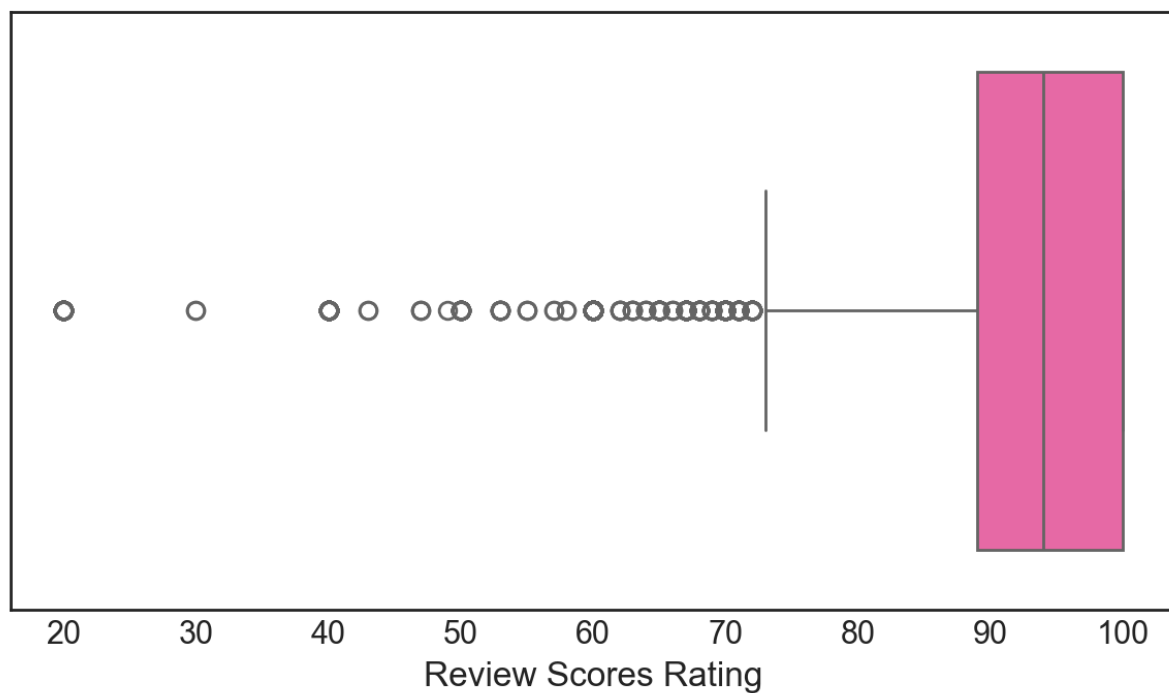
Box Plot of Price



Price had the most outliers which made uncovering the pattern of data harder, and made the visualizations trickier.



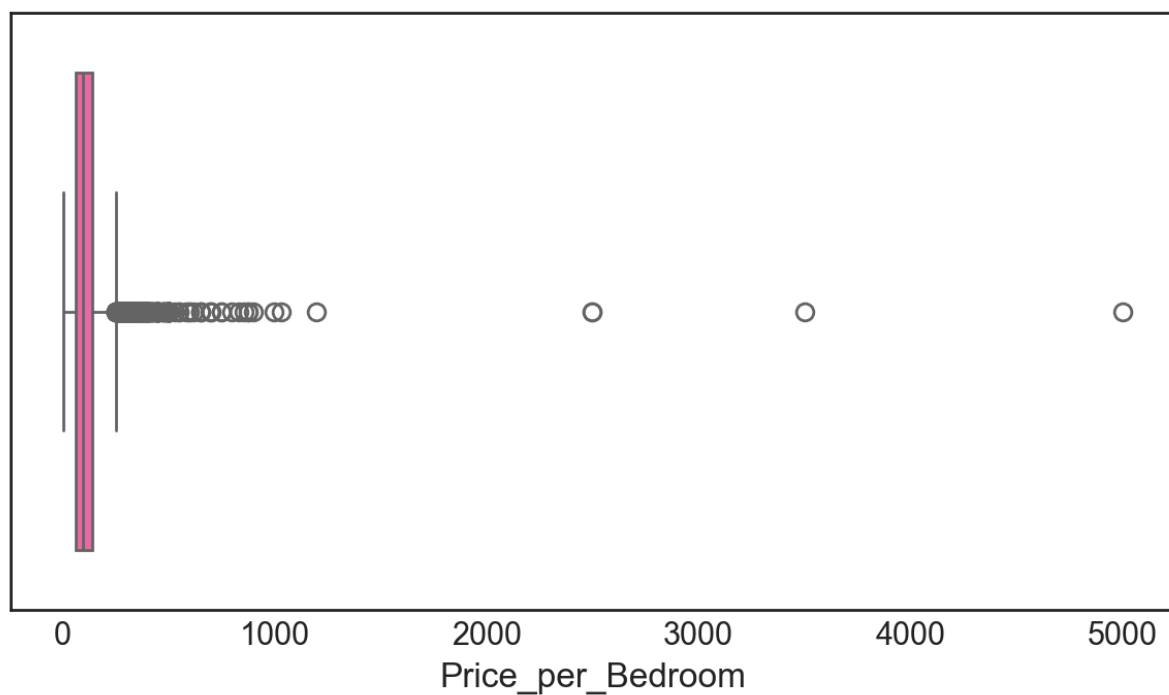
Box Plot of Review Scores Rating



Review Score Ratings was the only attribute that had outliers on its left side (tail). Its median is in the middle.



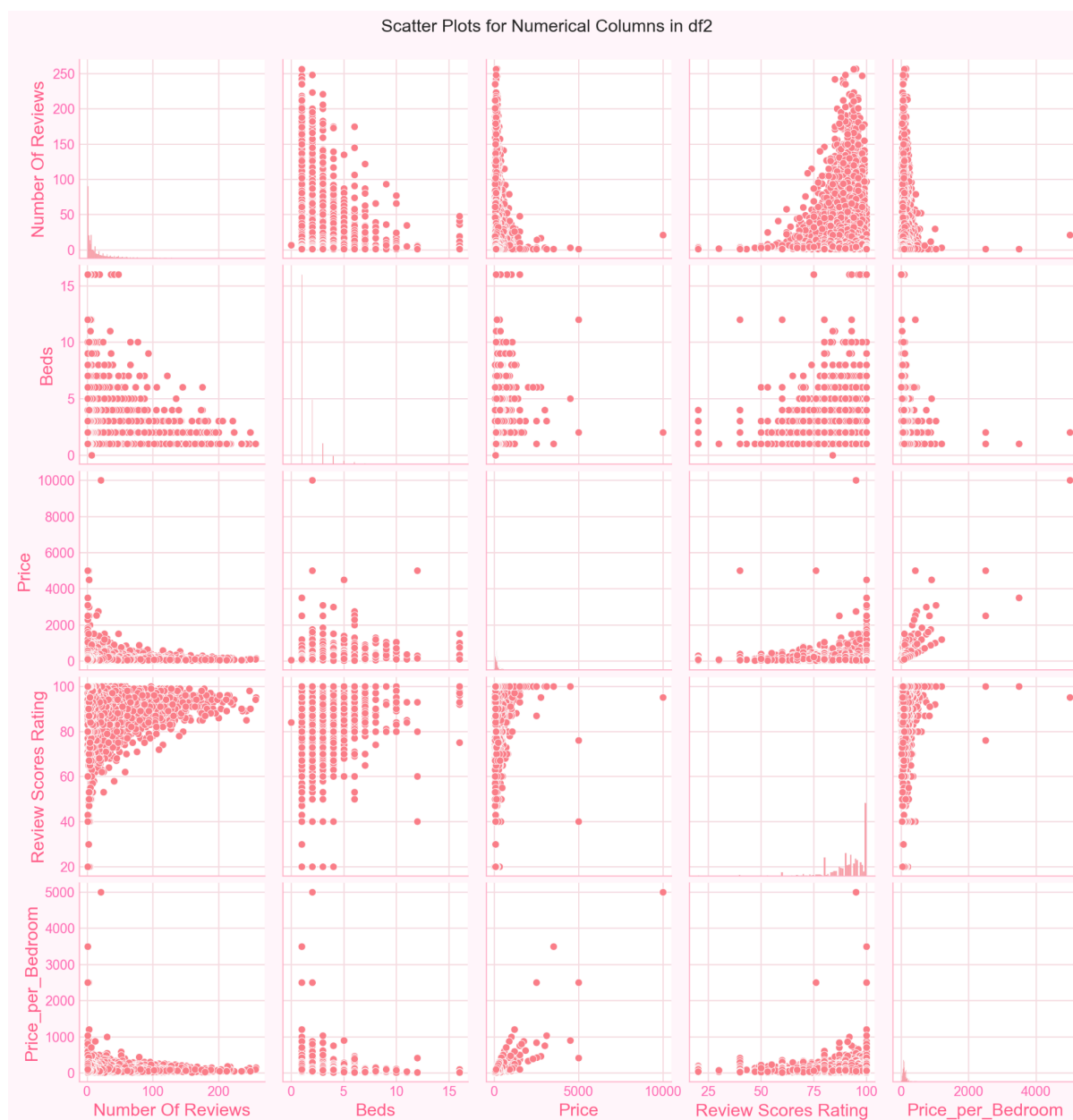
Box Plot of Price_per_Bedroom



Price per bedroom is negatively skewed and had a lot of outliers, the biggest at 5000 dollars.

Bivariate Analysis

Scatter plots



It shows that number of reviews had no correlation to price per bedroom, price or review score ratings. Also number of bedrooms had no correlation with review score ratings or price



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per bedroom. However price and price per bedroom had the most correlation. And higher prices also correlated with higher review ratings.

Correlation Matrix

A table that showed the correlation between numerical attributes from 1 (highly positively correlated) to -1 (highly negatively correlated). Price per bedroom and price were positively correlated. Number of bedrooms and ratings had no correlation.

Barcharts





This graph showed that the room type entire house/apt had the highest price per bedroom while a shared room had the lowest.

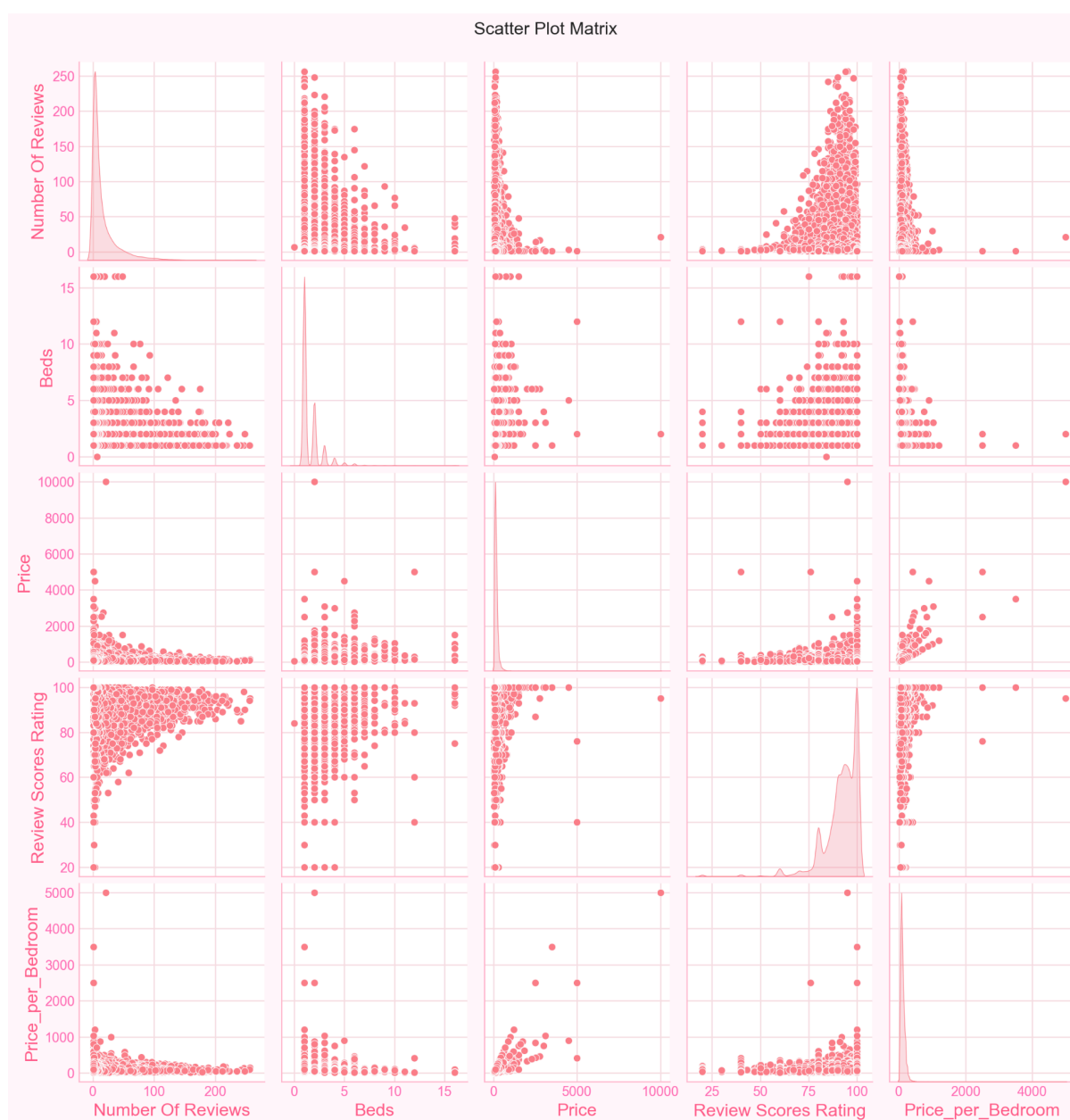
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This bar graph showed the price per bedroom compared to property types. RVs and Tents were the least expensive, while Other, Castle and Cabins were the most expensive per bedroom.

Multivariate Analysis

Scatterplot Matrix

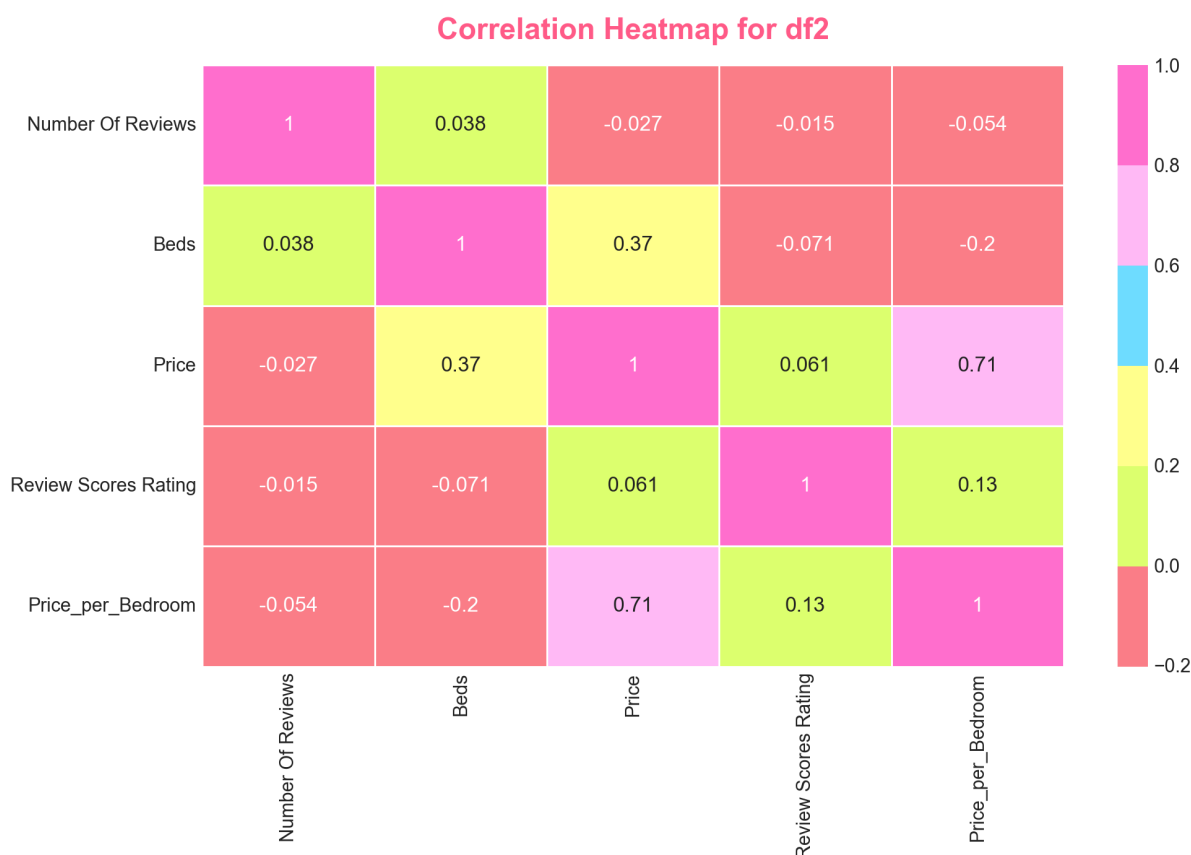


Same as above.



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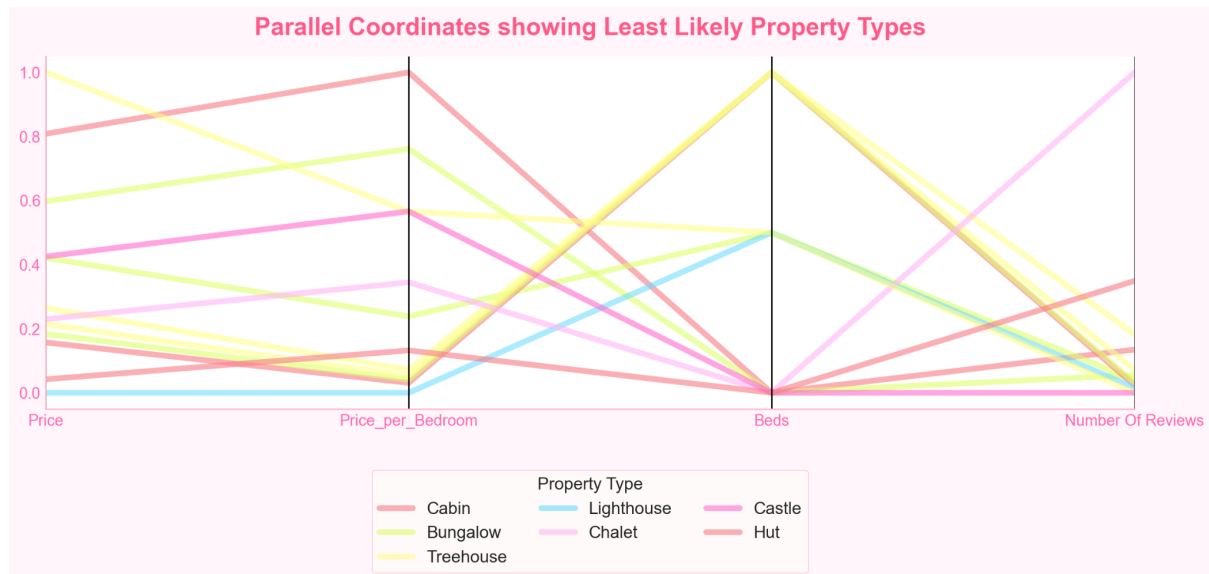
Heatmap



It shows that number of reviews had no correlation to price per bedroom, price or review score ratings. Also number of bedrooms had no correlation with review score ratings or price per bedroom. However price and price per bedroom had the most correlation. And higher prices also correlated with higher review ratings.

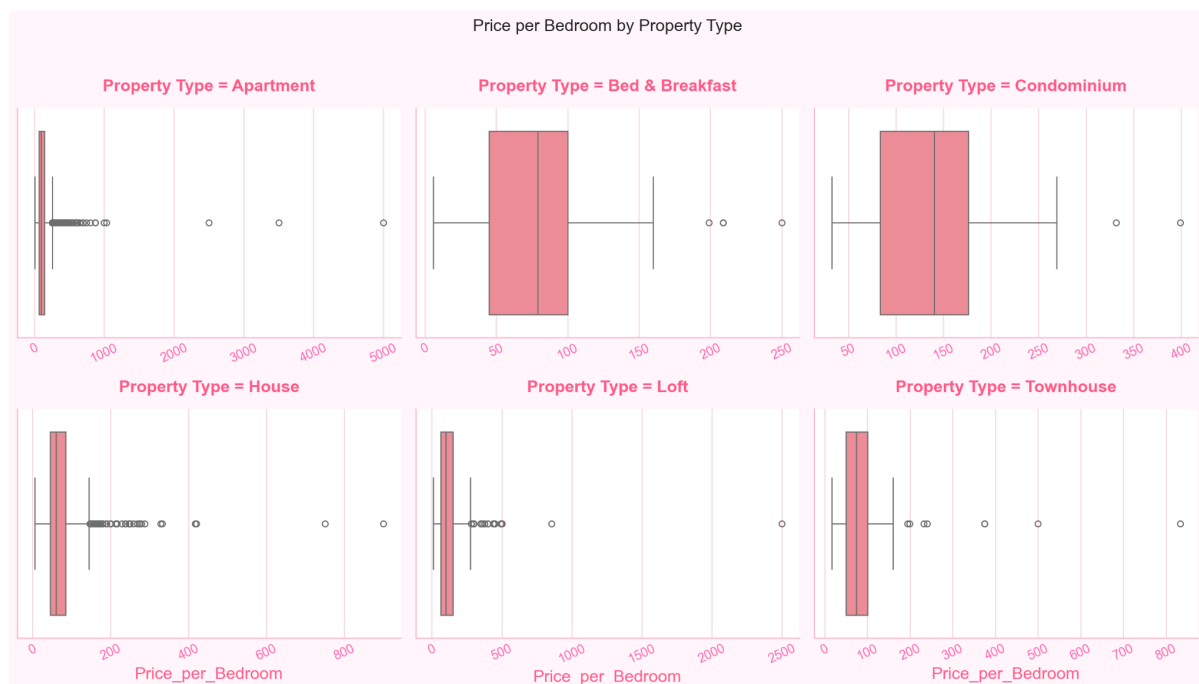


Parallel Coordinates Plot



The graph shows the 8 least common property types compared to numeric columns price, price per bedroom and beds and number of reviews. The axis is normalized to be the same for all. The number of reviews are the highest for chalet which has no beds. Price and price per bedroom are positively correlated. Most property types have close to zero reviews.

Trellis Plot



The trellis plot shows the distribution of the 6 most common property types by their price per bedroom. All of them have outliers greater than their Q3s. Bed and Breakfast and Condominiums are the most expensive on average. Apartments are the least expensive on average.

Bonus: Chernoff Faces



As a bonus I made these funny faces that capture 9 numerical columns using the dictionary:



```
feature_mapping = {  
    'Upper Face Height': all_numerical_cols_df2[0],  
    'Lower Face Overlap': all_numerical_cols_df2[1],  
    'Vertical Face Size': all_numerical_cols_df2[2],  
    'Upper Face Width': all_numerical_cols_df2[3],  
    'Lower Face Width': all_numerical_cols_df2[4],  
    'Nose Length': all_numerical_cols_df2[5],  
    'Mouth Position': all_numerical_cols_df2[6],  
    'Mouth Curvature': all_numerical_cols_df2[7],  
    'Mouth Width': all_numerical_cols_df2[8]  
}
```

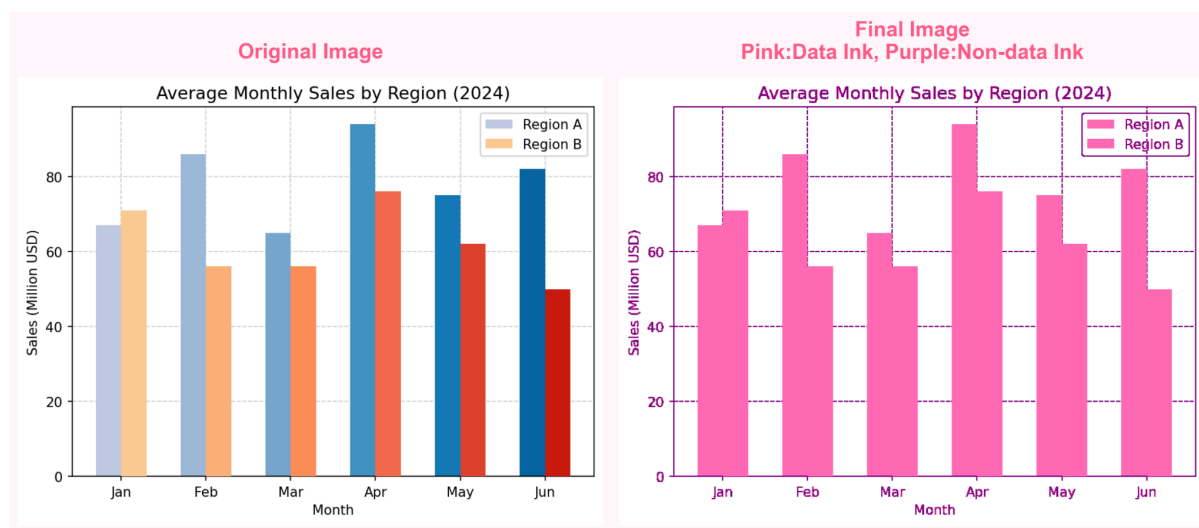
The fun part is that I pick any random 4 rows each time so they look different each time.



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Data Ink

Image 1

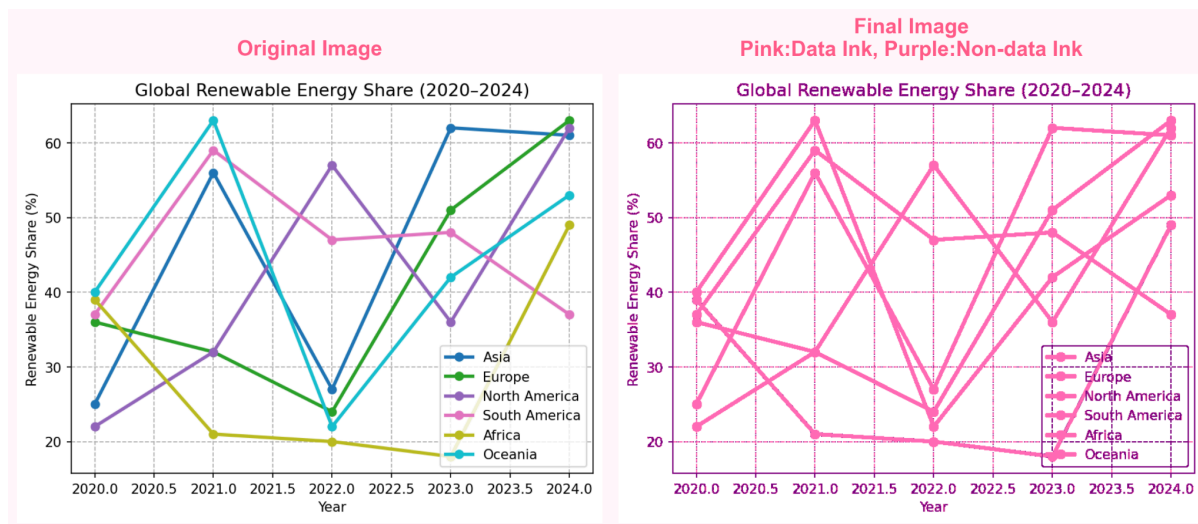


Data Ink Pixels: 197037

Non-Data Ink Pixels: 443889

Data Ink Ratio: 30.74%

Image 2

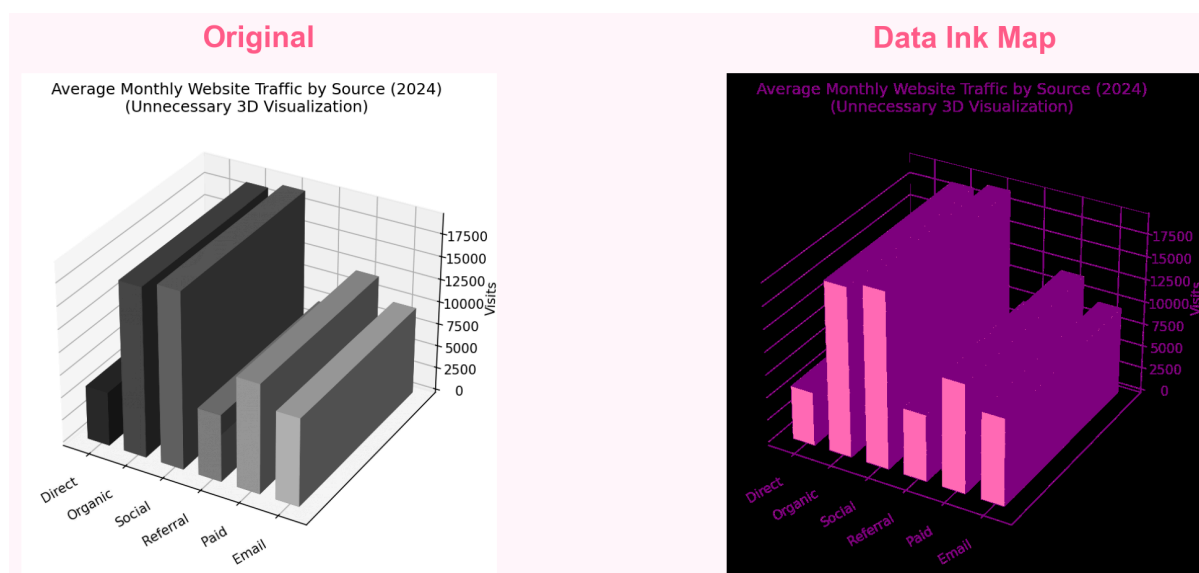


Data Ink Pixels: 59959

Non-Data Ink Pixels: 580967

Data Ink Ratio: 9.36%

Image 3



Data Ink Pixels: 38414

Non-Data Ink Pixels: 158312

Data Ink Ratio: 19.53%